Solution: Simplify Path

Problem Statement

Given an absolute file path in a Unix-style file system, simplify it by converting ".." to the previous directory and removing any "." or multiple slashes. The resulting string should represent the shortest absolute path.

Examples:

```
1.
   Input: "/a//b////c/d//././/.."
   Output: "/a/b/c"

2.
   Input: "/../"
   Output: "/"

3.
   Input: "/home//foo/"
   Output: "/home/foo"
```

Solution

To simplify the path, we'll use a stack to track the directories we're currently in. We'll split the input path into components by the "/" character, then process each component one by one. If a component is "..", we go to the previous directory by popping the top of the stack. If a component is "." or an empty string, we do nothing. Otherwise, we push the component into the stack as a new directory.

Detailed Algorithm steps:

1. Split the input path by the "/" character into an array of components.
2. Initialize an empty stack.
3. For each component in the array:
    - If the component is "..", pop the top of the stack (if it's not already empty).

- Else if the component is "." or an empty string, do nothing.
- Else, push the component into the stack as a new directory.

4. Finally, combine the components in the stack into a string, separated by the "/" character. Add a "/" at the start to denote an absolute path.

Algorithm walkthrough

let's walk through the code using the input `"/a//b////c/d//././/.."` step by step:

1. **Initialize a Stack**: A `Stack<String>` is created to store components of the simplified path.

2. **Split the Path**: The input path `"/a//b////c/d//././/.."` is split using `"/"` as the delimiter. The resulting parts are: `["", "a", "", "b", "", "", "", "c", "d", "", ".", "", ".", "", "", ".."]`. Empty strings and dots (`"."`) represent the current directory and will be ignored in further processing.

3. **Process Each Part**:

   - **First Part** (`""`): It's empty, so it's ignored.
   - **Second Part** (`"a"`): It's a directory name and is pushed onto the stack.
   - **Third Part** (`""`): Ignored.
   - **Fourth Part** (`"b"`): Another directory name, pushed onto the stack.
   - **Next Several Parts** (`""`, `""`, `""`): All empty, so ignored.
   - **Part** `"c"`: A directory name, pushed onto the stack.
   - **Part** `"d"`: Another directory name, pushed onto the stack.
   - **Part** `"."`: Represents the current directory, ignored.
   - **Part** `"."` (**again**): Again, represents the current directory, ignored.
   - **Last Part** (`".."`): Represents moving up one directory. It pops `"d"` from the stack.

   After processing, the stack contains: `["a", "b", "c"]`.

4. **Reconstruct Simplified Path**:

   - A `StringBuilder` is used to construct the final simplified path.
   - The stack is processed in LIFO (Last-In-First-Out) order. Each component is popped and appended to the start of the result string, preceded by `"/"`.
   - After processing the stack, the `StringBuilder` contains `"/a/b/c"`.

5. **Return the Result**:

- The `StringBuilder` is converted to a string and returned.
- For the given input, the output is `"/a/b/c"`.

Here is the visual representation of the algorithm:

Example 1:

Input: /a//b////c/d//././/..

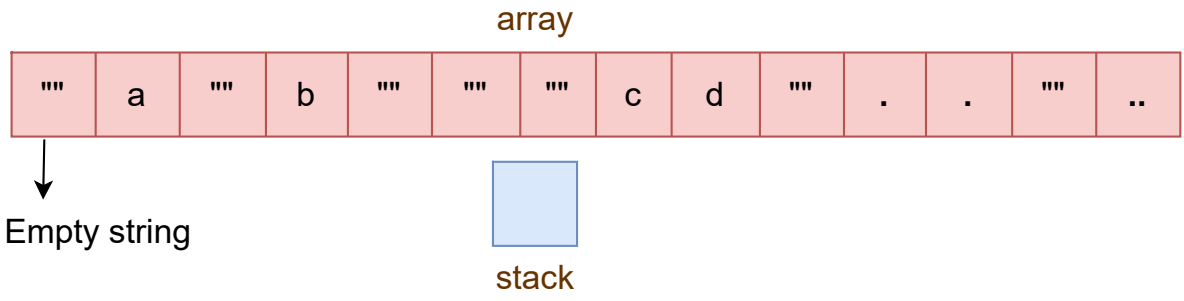
1. Initialize an empty stack to track the directories we're currently in.



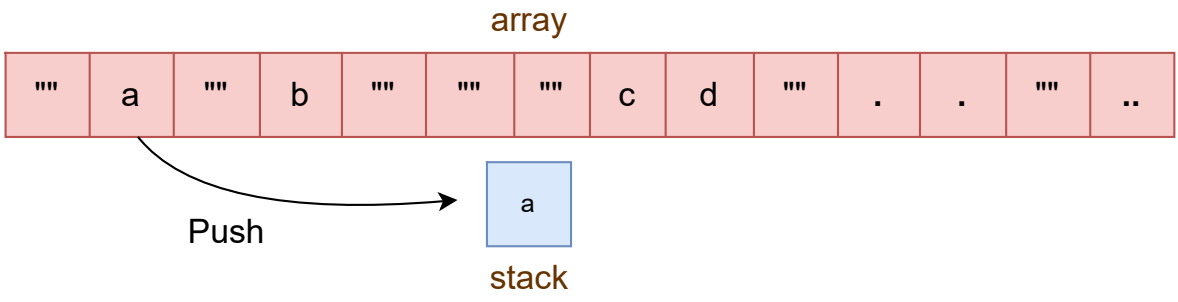
2. Split the input path into components by the "/" character, then process each component one by one.

3. If a component is "..", we go to the previous directory by popping the top of the stack.

4. If a component is "." or an empty string, we do nothing else push the component into the stack as a new directory.

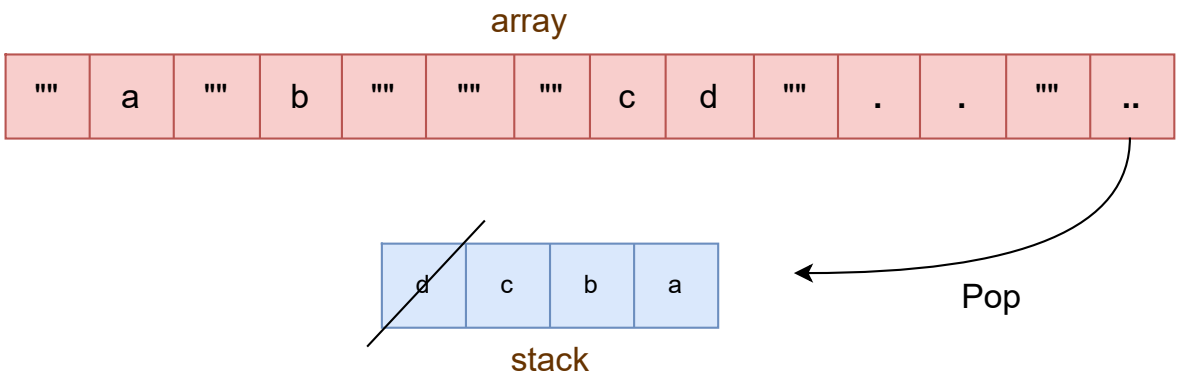


Don't push anything.



Push "a".

Repeat the same steps till we reach last element.



Pop top element "d"

Pop top element "d".

4. Combine the components in the stack into a string. Add a "/" at the start to denote an absolute path.

/a/b/c

Make the string in such a that way that the characters represents their positions from the original string.

Output: /a/b/c

Simplify Path

Code

Here is the code for this algorithm:

Java

```java
import java.util.*;

public class Solution {
    public String simplifyPath(String path) {
        // Create a stack to store the simplified path components
        Stack<String> stack = new Stack<>();

        // Split the input path string using '/' as a delimiter
        for (String p : path.split("/")) {
            if (p.equals("..")) {
                // If the component is '..', pop the last component from the stack
                if (!stack.isEmpty()) {
                    stack.pop();
                }
            } else if (!p.isEmpty() && !p.equals(".")) {
                // If the component is not empty and not '.', push it onto the stack
                stack.push(p);
            }
        }

        // Create a StringBuilder to build the simplified path
        StringBuilder result = new StringBuilder();

        // Reconstruct the simplified path by ...ping components from the stack
```

Run Code Result | Stdout

You must run your code first

Execute

Time and Space Complexity

The time complexity of the algorithm is $O(n)$, where n is the size of the input path, since we process each character once. The space complexity is also $O(n)$, as we store each directory in a stack.

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Introduction To Monotonic Stack

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